

Research Article

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Hirotake Yaguchi*

Generation of nonrecursive n -bit pseudorandom numbers based on β -transformation on $[1, 2)$ ($n = 64, 128, 192, \dots, 8192$)

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Abstract: We show that we can generate nonrecursive n -bit pseudorandom numbers using a simple algorithm whose essential computation is five times repetition of $(n\text{-bit}) \times (n\text{-bit})$ multiplication and taking out an n -bit integer from the result of multiplication. The algorithm can be described by β -transformation

$$T_{\beta}(X) = \beta X - \lfloor \beta X \rfloor + 1, \quad X \in [1, 2), \beta > 1.$$

We consider the condition that repetition of β -transformation generates random numbers, and see why our simple algorithm works well for various values of n

Keywords: Random number, pseudorandom number generation, nonrecursive, β -transformation

MSC 2020: 65C10, 11K45, 37E05

1 Introduction

Since we constructed a sequence of nonrecursive random numbers (RNs) from the chaotic behavior encountered when computing $\sin x = x - \frac{x^3}{3} + \dots$ numerically [5], we tried to find an underlying mechanism which causes the randomness and has mathematical background [2–4, 6]. Now we have reached a simple and general algorithm of generating nonrecursive n -bit RNs which works well for $n = 64, 128, 192, 256, 384, 512, 1024, 2048, 4096, 8192$ (and perhaps arbitrarily large n).

In the next section, we give the algorithm of generating RNs, which we call xMM, and then test randomness of generated RNs. The test proceeds using Sts-2.1.2 of NIST [7] and TestU01 by L'Ecuyer & Simard [1]. In Section 3, we show that the algorithm xMM is described by five times repetition of the fixed-bit-length discrete β -transformation on $[1, 2)$. In Section 4, by introducing the unfixed-bit-length discrete β -transformation, it is shown that we can expect help from the theory of β -transformation in ergodic theory. In Section 5, by changing the times of repetition of fixed-bit-length β -transformation, we confirm empirically that the suitable minimum number of repetitions of xMM is 5. In Section 6, by modifying the algorithm of xMM a little, we discuss the relation between the times of repetition of β -transformation and the randomness. In the last section, we consider mathematically a condition that repetition of β -transformation generates RNs, and see the reason why our xMM works well regardless of the value of n .

*Corresponding author: Hirotake Yaguchi, Ajina 4-32-7, Hatsukaichi City, Hiroshima Prefecture, 738-0054, Japan,
e-mail: hiroshimayaguchi@yahoo.co.jp

2 Generation and test of nonrecursive n -bit random numbers

2.1 Generation of random numbers (xMM)

We generate nonrecursive random numbers (RNs) using the following simple algorithm.

(A0) Prepare two n -bit integers B and x . We assume the MSB (most significant bit) of B is 1.

(A1) Set the MSB of x to be 1 and write X .

(A2) Multiply X by B , and obtain a $2n$ -bit number $\gamma \equiv B \cdot X$.

(A3) Take out n bits from just the middle of γ as the next x .

(A4) Repeat from (A1) to (A3) 5 times (we write $L_p = 5$), and use the last x as a random number $z (\equiv x)$.

(A5) To generate a sequence of random numbers $z_0, z_1, z_2, \dots$, nonrecursively, vary B in (A0) as $B_0, B_1, B_2, \dots$, and apply (A0)–(A4) to each B_i .

(We call the algorithm stated above **xMM** (extract Middle and Multiply) for convenience sake.)

The n -bit integers B and x in (A0) are defined as follows. Let e be Napier's constant $2.7182\dots$. The hexadecimal representation of the number $(1 + \frac{e}{10}) \times 8 = 10.1746254\dots$ is `0xa.2cb4411ba257552232c1139a172a5c9...` (0x means hexadecimal). Then, as x , we use the leading (first) n bits of the number `0xa.2cb4411...`. For example, if $n = 64$, then $x = \text{0xa2cb4411ba257552}$. (Note that the number x in (A0) is fixed throughout process (A5).) To define B_i in (A5), we use x in (A0) and i . That is, we represent i in 64 bits, and take XOR (exclusive OR, $\oplus$) with x at just the middle of x . (In the case of $n = 64$, i is represented in 48 bits.) For example, if $n = 64$ and $i = 3$, then B_3 is

$$(\text{a2cb4411} : \text{ba257552}) \oplus (000000 : 000003) = (\text{a2cb4411} : \text{ba257652}).$$

Of course, one can define x and B_i in a different way. For example, we used $\pi - 2.5 = 0.64159\dots = \text{0x0.a43f6a} \dots$ as x in the case of $n = 8192$.

2.2 Test of randomness of generated numbers

Randomness of numbers $\{z_i\}_i$ generated in (A5) was tested by Sts-2.1.2 of NIST [7] and TestU01 by L'Ecuyer & Simard [1] in two ways: one is to test all bits of z_i (write [Obj:A] for short), and another is to test the leading 64 bits of z_i (write [Obj:B] for short).

Sts-2.1.2 of NIST

We applied Sts-2.1.2 to 1000 bit-streams consisting of 1000000 bits per stream. (The block length of the block frequency test was modified to 20000.) The result of Sts-2.1.2 consists of 188 pairs of p -value and proportion. We repeat the Sts suite three times, and if $\{z_i\}_i$ fails to pass the same test 2 or 3 times, we decide that $\{z_i\}_i$ cannot pass that test. Then the result was that all tests were passed for $n = 64, 128, 192, 256, 384, 512, 1024, 2048, 4096, 8192$ and for [Obj:A], [Obj:B].

TestU01 by L'Ecuyer & Simard

We applied the battery Crush for $n = 64, 128, 192, 256, 384, 512, 1024, 2048$ and 4096. For $n = 8192$, the battery SmallCrush was applied, considering the time required. Crush consists of 96 items of tests and generates 144 p -values. If the p -value of a test is out of the range $[0.001, 0.9990]$, we further apply additional tests of three times, and if two or three p -values are out of the range, we decide that $\{z_i\}_i$ cannot pass that test. We treat p -values of SmallCrush in the same way. Then the result was that all tests were passed for $n = 64, 128, 192, 256, 384, 512, 1024, 2048, 4096, 8192$ and for [Obj:A], [Obj:B] (see also Table 1). (In the case of $n = 4096$ for [Obj:B], the total time to complete the test was about one month.)

3 Description of xMM by fixed-bit-length β -transformation

Let T_β , $\beta > 1$, be the transformation on $[1, 2)$ defined by $T_\beta(X) = \beta X - \lfloor \beta X \rfloor + 1$, $X \in [1, 2)$, where $\lfloor q \rfloor$ is the largest integer not exceeding q . We call T_β the β -transformation on $[1, 2)$ following the terminology in ergodic theory. In Figure 1, we give examples of the graph of T_β and T_β^2 . (Symbols $\mathbf{F}$, $\mathbf{H}_1$ and $\mathbf{L}_1$ will be used in Section 7).

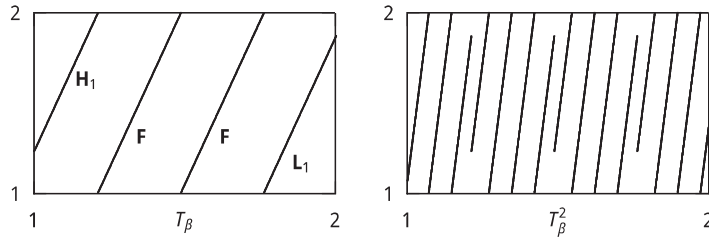


Figure 1: Examples of the graph of T_β and T_β^2

We here *always* assume that β has the form

$$\beta = 2^k B, \quad B \in [1, 2), \quad k: \text{positive integer.}$$

(We dare to use the same symbols B and X which are used in Section 2.1. The reason will be clear afterwards.)

Let the product BX in $\beta X \equiv 2^k BX$ be represented as

$$BX = w_0 w_1 . w_2 w_3 w_4 \dots$$

in binary ($w_0 w_1 = 01$ if $BX \in [1, 2)$, and $w_0 = 1$ if $BX \in [2, 4)$). Then $2^k BX$ is represented as

$$2^k BX = w_0 w_1 \dots w_{k+1} . w_{k+2} w_{k+3} \dots,$$

that is, BX is shifted to the left by k with respect to the (fixed) zero point,

$$BX = w_0 w_1 . w_2 w_3 \dots \xrightarrow{\text{shift left by } k} 2^k BX = w_0 w_1 \dots w_{k+1} . w_{k+2} \dots$$

The final value $T_\beta(X) \equiv T_{2^k B}(X)$ is obtained by discarding (dropping) $w_0 w_1 \dots w_k$ and replacing w_{k+1} with 1, that is,

$$T_\beta(X) \equiv T_{2^k B}(X) = 1 . w_{k+2} w_{k+3} \dots$$

Usually, in computer, integers are represented under a restricted bit length, say n -bit, and results of multiplication of two n -bit integers are represented in $2n$ -bit. Therefore, when computing $T_\beta(X)$ numerically, we follow this restriction. Let n -bit B , X and $2n$ -bit BX be represented as

$$B \equiv 1 . u_1 \dots u_{n-1}, \quad X \equiv 1 . v_1 \dots v_{n-1} \quad \text{and} \quad BX \equiv w_0 w_1 . w_2 w_3 \dots w_{2n-1}.$$

Then we have

$$2^k BX = w_0 \dots w_{k+1} . w_{k+2} \dots w_{2n-1} \quad \text{and} \quad T_{2^k B}(X) = 1 . w_{k+2} \dots w_{2n-1}.$$

Since $T_{2^k B}(X)$ is a $((2n-1)-k)$ -bit number, in order to use this value as the next X , we truncate $T_{2^k B}(X)$ from $((2n-1)-k)$ -bit to n -bit by cutting off the last $(n-1)-k$ bits including LSB (least significant bit),

$$T_\beta(X) = 1 . w_{k+2} \dots w_{2n-1} \xrightarrow{\text{truncate}} 1 . w_{k+2} \dots w_{k+n} \quad (\text{used as the next } X).$$

Notice that, in the truncation above, there occurs a so-called rounding error due to cut-off of bits. We call the process above to get the next n -bit number $X = 1 . w_{k+2} \dots w_{k+n}$ from the initial n -bit number $X = 1 . v_1 \dots v_{n-1}$ the **fixed-bit-length (discrete) β -transformation** (f.b.l. BT for short). Remember that, in f.b.l. BT, the leading $k+1$ bits $w_0 w_1 \dots w_k$ of BX are dropped by a shift to the left and the last $(n-1)-k$ bits $w_{k+n+1} \dots w_{2n-1}$ of BX are cut off.

Now let us consider the special case $k = (n/2) - 1$. In this case, the flow of computing f.b.l. BT is written as follows:

$$\begin{aligned}
 X = 1.v_1 \dots v_{n-1} &\xrightarrow{(A2)} BX \equiv w_0 w_1 w_2 \dots w_{(n/2)} w_{(n/2)+1} \dots w_{2n-1} \\
 &\xrightarrow{\text{shift left by } (n/2)-1 \text{ bit}} 2^{(n/2)-1} BX = w_0 w_1 \dots w_{(n/2)} w_{(n/2)+1} \dots w_{2n-1} \\
 &\xrightarrow{\text{drop the leading } n/2 \text{ bits}} w_{(n/2)} w_{(n/2)+1} \dots w_{2n-1} \\
 &\xrightarrow{\text{MSB}=1} T_{2^{(n/2)-1}B}(X) = 1.w_{(n/2)+1} \dots w_{(3n/2)-1} w_{(3n/2)} w_{(3n/2)+1} \dots w_{2n-1} \\
 &\xrightarrow{\text{cut off the last } n/2 \text{ bits}} 1.w_{(n/2)+1} w_{(n/2)+2} \dots w_{(3n/2)-1} \quad (\text{the next } n\text{-bit } X).
 \end{aligned}$$

Since the next n -bit number $1w_{(n/2)+1}w_{(n/2)+2} \dots w_{(3n/2)-1}$ is located in just the middle of $2n$ -bit number $w_0 w_1 w_2 \dots w_{(n/2)} w_{(n/2)+1} \dots w_{2n-1}$, it is easy to see that the flow of f.b.l. BT given above corresponds to (A2), (A3), (A1) of xMM, and that (A4) means to repeat f.b.l. BT five times. Thus we can say that xMM is described by repeating f.b.l. BT with $k = (n/2) - 1$ five times. (Note that the MSB $w_{(n/2)}$ in the last f.b.l. BT is used as it is.)

4 Unfixed-bit-length β -transformation and ergodic theory

The β -transformation which frequently appears in ergodic theory is of the form $\Phi_\beta(t) = \beta t - \lfloor \beta t \rfloor$, $t \in [0, 1)$, $\beta > 1$. The linear mod 1 transformation is an extension of Φ_β and is defined by $\Phi_{\beta, \alpha}(t) = \beta t + \alpha - \lfloor \beta t + \alpha \rfloor$, $t \in [0, 1)$, $\beta > 1$, $\alpha \in (0, 1)$. These two, which we call simply β -transformation, are transformations on $[0, 1)$. On the other hand, our β -transformation $T_\beta(X) = \beta X - \lfloor \beta X \rfloor + 1$, $X \in [1, 2)$, is defined on $[1, 2)$. However, $T_\beta(X)$ is deformed as

$$T_\beta(X) = \beta X - \lfloor \beta X \rfloor + 1 = \{\beta(X - 1) + (\beta - \lfloor \beta \rfloor)\} - \lfloor \beta(X - 1) + (\beta - \lfloor \beta \rfloor) \rfloor + 1,$$

and so T_β is obtained from $\Phi_{\beta, \beta - \lfloor \beta \rfloor}(t) = \{\beta t + (\beta - \lfloor \beta \rfloor)\} - \lfloor \beta t + (\beta - \lfloor \beta \rfloor) \rfloor$ by translating rightward by 1 and upward by 1. This implies that the randomness of our xMM might be analyzed with help from the theory of β -transformation. However, f.b.l. BT of Section 3 includes the operation cut-off (rounding), which is not familiar with the theory of β -transformation. So we here introduce uf.b.l. BT (unfixed-bit-length (discrete) β -transformation). This is the one which is obtained from f.b.l. BT by omitting the operation cut-off. Therefore, in uf.b.l. BT, the value $T_{2^{(n/2)-1}B}(X)$ obtained from X is used as the next X directly. Thus uf.b.l. BT is a (discrete) β -transformation itself. (Notice that if we repeat uf.b.l. BT, then the bit length of $T_{2^{(n/2)-1}B}(X)$ becomes longer and longer.) The uf.b.l. BT has its meaning if the behavior of randomness is at the same level as that of f.b.l. BT.

Let $\{\zeta_i\}_i$ be the sequence of n -bit numbers which is obtained from uf.b.l. BT in the same way as $\{z_i\}_i$ of f.b.l. BT. Then the result of applying TestU01 (Crush) to $\{\zeta_i\}_i$ for $n = 64, 128, 192, 256, 384, 512$ and for [Obj:A], [Obj:B] was that all tests were passed except only one test of $n = 64$ (see the row $L_p = 5$ in Table 1). Hence we can think that the randomness of $\{\zeta_i\}_i$ and $\{z_i\}_i$ are at the same level, and then we can expect help from the theory of β -transformation.

5 Relations between times of repetition of xMM and randomness

The times of repetition $L_p = 5$ in xMM are determined empirically by changing the value of L_p variously. Table 1 shows the number of tests in TestU01 (Crush) whose p -values were out of the range $[10^{-15}, 1 - 10^{-15}]$. More precisely, $\uparrow e$ (resp. $\uparrow e, \downarrow e, \downarrow e$) is the number of tests in Crush whose p -values were in $(1 - 10^{-15}, 1 - 10^{-300}]$ (resp. $(1 - 10^{-300}, 1)$, $[10^{-300}, 10^{-15})$, $(0, 10^{-300})$), and the symbol o means that all p -values were in $[10^{-15}, 1 - 10^{-15}]$. From the table, we know that $L_p = 5$ is the reasonable minimum value for f.b.l. BT to generate RNs. Notice that, in the case of $L_p = 5$ (xMM), the total bits which are dropped in the repetition of f.b.l. BT are $\frac{n}{2} \times 5 = (5/2)n$ bits regardless of the value of bit length n .

xMM	$n = 64$		$n = 128$		$n = 192$		$n = 256$	
	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT
$L_p = 3$	$\uparrow 3 \downarrow 4 \Downarrow 27$	$\uparrow 4 \downarrow 3 \Downarrow 39$	$\uparrow 2 \downarrow 5 \Downarrow 34$	$\uparrow 4 \downarrow 3 \Downarrow 40$	$\downarrow 4 \Downarrow 33$	$\uparrow 4 \downarrow 3 \Downarrow 34$	$\uparrow 4 \downarrow 1 \Downarrow 40$	$\uparrow 4 \downarrow 1 \Downarrow 41$
$L_p = 4$	$\Downarrow 2$	$\downarrow 4 \Downarrow 14$	$\downarrow 1 \Downarrow 3$	$\uparrow 3 \downarrow 2 \Downarrow 14$	o	o	o	$\downarrow 2 \Downarrow 2$
$L_p = 5$	o	$\Downarrow 1$	o	o	o	o	o	o
$L_p = 6$	o	o	o	o	o	o	o	o

Table 1

6 Relations between dropping the first $4/n$ bits and randomness

We consider here the case that (differently from xMM) the k of $T_{2^k B}$ is $\frac{n}{4} - 1$. This is the case that the bits dropped in f(uf).b.l. BT are leading $n/4$ bits. Table 2 is made in the same way as in Section 5.

Drop $4/n$	$n = 64$		$n = 128$		$n = 192$		$n = 256$	
	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT	f.b.l. BT	uf.b.l. BT
$L_p = 6$	$\uparrow 1 \downarrow 2 \Downarrow 11$	$\uparrow 3 \downarrow 4 \Downarrow 24$	$\downarrow 2 \Downarrow 7$	$\downarrow 3 \Downarrow 18$	$\downarrow 1 \Downarrow 11$	$\uparrow 1 \downarrow 8 \Downarrow 21$	$\uparrow 2 \downarrow 4 \Downarrow 23$	$\uparrow 2 \downarrow 2 \Downarrow 30$
$L_p = 7$	$\Downarrow 3$	$\downarrow 1 \Downarrow 4$	$\Downarrow 1$	$\downarrow 1 \Downarrow 5$	$\Downarrow 1$	$\Downarrow 2$	$\downarrow 1 \Downarrow 1$	$\uparrow 1 \downarrow 3 \Downarrow 9$
$L_p = 8$	$\Downarrow 1$	$\Downarrow 2$	o	$\Downarrow 1$	o	o	o	$\Downarrow 1$
$L_p = 9, 10$	o	o	o	o	o	o	o	o

Table 2

From Table 2, we can think that randomness appears from $L_p = 9$ regardless of the value of n . Notice that, in this case, since the first $n/4$ bits are dropped in f(uf).b.l. BT, a total of $\frac{n}{4} \times 9 = (9/4)n$ bits are dropped regardless of the value of bit length n .

7 Mathematical consideration

In this section, we consider the reason why $(5/2)n$ in Section 5 and $(9/4)n$ in Section 6 appear as the total bits to be dropped to get RNs. Let I_m be the integer function defined by

$$T_{2^k B}^m(X) = T_{2^k B}(T_{2^k B}^{m-1}(X)) = 2^k B \cdot T_{2^k B}^{m-1}(X) - I_m, \quad I_m = \lfloor 2^k B \cdot T_{2^k B}^{m-1}(X) \rfloor - 1.$$

Then we have

$$T_{2^k B}^\ell(X) = (2^k B)^\ell X - (2^k B)^{\ell-1} I_1 - (2^k B)^{\ell-2} I_2 - \cdots - (2^k B) I_{\ell-1} - I_\ell.$$

($X = I_1 + I_2(2^k B)^{-1} + I_2(2^k B)^{-2} + \cdots$ is called the β -expansion of X .)

We focus on the coefficient $(2^k B)^\ell$ of X in the right-hand side. The number $(2^k B)^\ell$, $2^k \gg \ell$, approximates the number of lines (branches) which appear in the graph of $T_{2^k B}^\ell([1, 2))$. More precisely, it holds that

$$(1 - 2^{-k+1})^\ell \leq \frac{\#(T_{2^k B}^\ell)}{(2^k B)^\ell} \leq (1 + 2^{-k+1})^\ell,$$

where $\#(T_{2^k B}^\ell)$ denotes the number of lines in the graph of $T_{2^k B}^\ell([1, 2))$. The proof is elementary. In the graph of $T_\beta([1, 2))$, there exist three kinds of lines (branches) **F**, **H**₁ and **L**₁ (Figure 1), and it is clear that the number of **F**s is $\lfloor 2\beta \rfloor - \lfloor \beta \rfloor - 1$. A lower bound of $\#(T_\beta^1)$ is $\lfloor 2\beta \rfloor - \lfloor \beta \rfloor - 1$, which is the number of lines **F** in the graph. An upper bound of $\#(T_\beta^1)$ is $\lfloor 2\beta \rfloor - \lfloor \beta \rfloor + 1$, which is the number of lines in the graph. Hence we have $\lfloor 2\beta \rfloor - \lfloor \beta \rfloor - 1 \leq \#(T_\beta^1) \leq \lfloor 2\beta \rfloor - \lfloor \beta \rfloor + 1$. In the next step $\#(T_\beta^2)$, it is clear that $(\lfloor 2\beta \rfloor - \lfloor \beta \rfloor - 1)^2$ is a lower bound.

Thinking that there are $\lfloor 2\beta \rfloor - \lfloor \beta \rfloor + 1$ Fs in a graph by replacing $\mathbf{H}_1$ and $\mathbf{L}_1$ with $\mathbf{F}$, we have an upper bound $(\lfloor 2\beta \rfloor - \lfloor \beta \rfloor + 1)^2$ of $\#(T_\beta^2)$. In this way, we obtain

$$(\lfloor 2\beta \rfloor - \lfloor \beta \rfloor - 1)^\ell \leq \#(T_\beta^\ell) \leq (\lfloor 2\beta \rfloor - \lfloor \beta \rfloor + 1)^\ell.$$

Now let $\hat{\beta} = \beta - \lfloor \beta \rfloor$, and so $\beta = \lfloor \beta \rfloor + \hat{\beta}$, $\hat{\beta} \in [0, 1)$. Then, noting that $\lfloor \hat{\beta} \rfloor = 0$, the inequality is rewritten as

$$(\lfloor \beta \rfloor + \lfloor 2\hat{\beta} \rfloor - 1)^\ell \leq \#(T_\beta^\ell) \leq (\lfloor \beta \rfloor + \lfloor 2\hat{\beta} \rfloor + 1)^\ell.$$

In the case of $\beta = 2^k B$, since $\lfloor \beta \rfloor = 2^k B - \hat{\beta}$, we have

$$(2^k B - \hat{\beta} + \lfloor 2\hat{\beta} \rfloor - 1)^\ell \leq \#(T_\beta^\ell) \leq (2^k B - \hat{\beta} + \lfloor 2\hat{\beta} \rfloor + 1)^\ell.$$

If we use $0 < \hat{\beta} < 1$ and $\lfloor 2\hat{\beta} \rfloor = 0$ or 1 , the inequality above yields $(2^k B - 2)^\ell \leq \#(T_\beta^\ell) \leq (2^k B + 2)^\ell$, that is,

$$\left(1 - \frac{2}{2^k B}\right)^\ell \leq \frac{\#(T_\beta^\ell)}{(2^k B)^\ell} \leq \left(1 + \frac{2}{2^k B}\right)^\ell.$$

Because $1 < B < 2$, we finally get

$$(1 - 2^{-k+1})^\ell \leq \frac{\#(T_\beta^\ell)}{(2^k B)^\ell} \leq (1 + 2^{-k+1})^\ell.$$

(Numbers of branches in the graph of T_β^ℓ were counted more precisely in [2] in connection with the invariant measure of T_β .)

The minimum step of $B = 1.u_1 \dots u_{n-1}$ is $\Delta \equiv 2^{-(n-1)}$, so the next neighboring value of $(2^k B)^\ell$ is $(2^k(B + \Delta))^\ell$. Then the difference of $(2^k B)^\ell$ and $(2^k(B + \Delta))^\ell$ is

$$(2^k(B + \Delta))^\ell - (2^k B)^\ell = 2^{k\ell} \{\ell B^{\ell-1} \Delta + \ell C_2 B^{\ell-2} \Delta^2 + \dots + \ell B \Delta^{\ell-1} + \Delta^\ell\}.$$

Since Δ is very small, the important part of the expansion above is the first term $\ell 2^{k\ell} B^{\ell-1} \Delta$, and the essential part of $\ell 2^{k\ell} B^{\ell-1} \Delta$ is $2^{k\ell} \Delta$. Let us consider $R(k, \ell) \equiv 2^{k\ell} \Delta / 2^n = 2^{k\ell - (2n-1)}$, which is a kind of normalization of $2^{k\ell} \Delta$ by the bit length n . We are interested in the case that $k = \frac{n}{s} - 1$, where the bits dropped in f(uf).b.l. BT are $\frac{n}{s}$ bits. In this case, $R(k, \ell)$ is rewritten as $R(\frac{n}{s} - 1, \ell) = 2^{(n/s-1)\ell - (2n-1)} = 2^{(\ell/s-2)n-\ell+1}$, and so we have

$$R\left(\frac{n}{s} - 1, 1\right) = 2^{(1/s-2)n}, \quad \dots, \quad R\left(\frac{n}{s} - 1, s\right) = 2^{-n-s+1}, \quad \dots, \quad R\left(\frac{n}{s} - 1, 2s\right) = 2^{-2s+1}.$$

From these values, we know that $\ell = 2s$ is an important turning point where the sign of n in the exponent of $2^{(\ell/s-2)n-\ell+1}$ changes from $-$ (minus) to $+$ (plus). Further, at $\ell = 2s$, the total bits dropped (t.b.dr for short) in the repetition of f(uf).b.l. BT are $\frac{n}{s} \times 2s = 2n$ bits. If we look at the results of Section 5 (t.b.dr = $2.5n$) and Section 6 (t.b.dr = $2.25n$), it looks like that, after the point where t.b.dr = $2n$, the randomness appears apparently.

As a result, we consider that, when n -bit RNs are generated by f(uf).b.l. BT under the condition that the leading $K(\leq \frac{n}{2})$ bits are dropped and the times of repetition is L_p , then K and L_p satisfy $K \times L_p > 2n$. If $K = \frac{n}{s}$ ($s = 2$ in Section 5 and $s = 4$ in Section 6), the inequality is reduced to $L_p > 2s$, which is independent of n . Especially in the case of $s = 2$ and $L_p = 5$, which satisfy $L_p > 2s$, the t.b.dr is $(5/2)n$ of Section 5, and the algorithm corresponding to f.b.l. BT is xMM in Section 2.1. Of course, t.b.dr = $(9/4)n$ of Section 6 is the case of $s = 4$ and $L_p = 9$ satisfying $L_p > 2s$.

Remark 1. Notice that the condition $L_p > 2s$ does not always imply the randomness. For example, in the case of $n = 64$ and $L_p = 33 > 32 = 2s$ ($s = 16$) where $n/s (= 4)$ is too small, the sequence of numbers generated by f.b.l. BT does not pass TestU01 (Crush). However, the following procedure improves the randomness remarkably[3]: after the last multiplication $B_i \times X$ in the loop of f.b.l. BT, drop the leading $32 (= \frac{n}{2})$ bits of $B_i \cdot X$ (instead of dropping $4 (= \frac{n}{s})$ bits), and take out the succeeding $64 (= n)$ bits as a random number. If we adopt this improvement, the sequence of numbers obtained passes TestU01 even if $L_p = 20$. This improvement suggested to us a way to xMM where the leading $n/2$ bits are dropped every time.

Remark 2. We have obtained mathematically a global condition of generating RNs as $L_p > 2s$. However, we do not have a good (mathematical) answer to the question of what is the origin of the randomness of xMM.

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